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Piston type pump drive arrangement

Abstract

A piston type pump includes a pump housing having a pump inlet, a pump outlet, and a piston arrangement connected to a drive shaft. The drive shaft is configured to drive the piston arrangement, and the drive shaft includes a first eccentric. The piston arrangement has a first primary stage piston connected to a first sliding block guide slidably seated on the first eccentric. The first sliding block guide has a main axis, a minor axis, and an inner surface. A first sliding bush is arranged between the first eccentric and the first sliding block guide. The outer surface of the first sliding bush corresponds to the inner surface of the first sliding block guide. Translational movement of the first sliding bush relative to the first sliding block guide is allowed along the main axis and restricted along the minor axis.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. National Phase application under 35 U.S.C. § 371 of International Application No. PCT/EP2019/063110, filed on May 21, 2019. The International Application was published in English on Nov. 26, 2020 as WO/2020/233796 A1 under PCT Article 21(2).

FIELD

(2) The present disclosure relates to a piston type pump, comprising a pump housing having at least one pump inlet and one pump outlet and a piston arrangement connected to a drive shaft which, when driven, sets the piston arrangement into movement, wherein the drive shaft comprises a first eccentric, and the piston arrangement comprises a first primary stage piston connected to a first sliding block guide slidably seated on the first eccentric, said first sliding block guide having a main axis, a minor axis and an inner surface. Such pumps may be used to induce a vacuum at the pump inlet and/or to provide pressurized fluid at the pump outlet.

BACKGROUND

(3) Vacuum pumps are for example known from WO 2017/137144 A1 or WO 2017/137141 A1. Such vacuum pumps are generally referred as piston type vacuum pumps in distinction from so-called rotary vane vacuum pumps. Piston type pumps include at least one piston which reciprocatingly moves inside a cylinder. The pump inlet usually is connected with the working chamber formed by the cylinder such that when the piston moves inside the cylinder for increasing the working volume of the working chamber the vacuum is induced at the inlet. A first piston rod of a first piston is driven by a drive shaft of the pump and performs a combined linear and pendulum motion. A second piston rod of the pumps is rotatable connected to the first piston rod via a connecting bolt and thus also performs a combination of a linear and a pendulum motion. However, the drive assembly of such pumps requires close manufacturing tolerances and comprises multiple parts, resulting in increased time for assembly and increased manufacturing cost. Furthermore, the non-rectilinear movement of the pistons might lead to increased wear on the piston and/or cylinder of the pump. Pumps of this type are used in passenger vehicles or trucks as in particular vacuum pumps to supply specific modules of the vehicle with a vacuum. Generally manufacturing cost and good wear characteristics are of utmost importance for pumps used in vehicles.

(4) A drive assembly of a compressed fluid motor, which provides a linear piston motion is presented in US 2017/0350249 A1. Pistons of said motor drive an output shaft via a crank pin. A rolling bearing positioned on the crank pin is slidably supported in between guide plates connected to respective piston rods of the pistons. During operation of said motor, the crank pin is rotated through a linear movement of the guide plates, wherein the outer ring of said rolling bearing slides relative to the guide plates and thereby eliminating a rocking motion of the piston rods. However, the presented drive assembly still comprises multiple parts requiring close manufacturing tolerances and therefore increasing a required time for assembly as well as manufacturing cost. Furthermore, the pistons are not fixed rotationally to the crank pin which can lead to irregular wear on piston seals. Moreover, drive assemblies are known, wherein a crank pin directly interacts with a guide slot without a bearing. While such assemblies may have fewer components highly accurate machining of the parts is necessary, in order to provide acceptable vibrational and noise levels.

SUMMARY

(5) In an embodiment, the present disclosure provides a piston type pump. The piston type pump includes a pump housing having a pump inlet, a pump outlet, and a piston arrangement connected to a drive shaft. The drive shaft is configured to drive the piston arrangement, and the drive shaft

includes a first eccentric. The piston arrangement has a first primary stage piston connected to a first sliding block guide slidably seated on the first eccentric. The first sliding block guide has a main axis, a minor axis, and an inner surface. A first sliding bush is arranged between the first eccentric and the first sliding block guide. The outer surface of the first sliding bush corresponds to the inner surface of the first sliding block guide. Translational movement of the first sliding bush relative to the first sliding block guide is allowed along the main axis and restricted along the minor axis.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Subject matter of the present disclosure will be described in even greater detail below based on the exemplary figures. All features described and/or illustrated herein can be used alone or combined in different combinations. The features and advantages of various embodiments will become apparent by reading the following detailed description with reference to the attached drawings, which illustrate the following:
(2) FIG. **1** shows a perspective simplified cut view of a piston type pump comprising an eccentric and sliding block guide drive arrangement;
(3) FIG. **2** shows a simplified perspective view of a piston arrangement of a piston type pump comprising an eccentric and sliding block guide drive arrangement;
(4) FIG. **3** shows a perspective view of a second primary stage piston and a first secondary stage piston attached together;
(5) FIG. **4** shows the arrangement of FIG. **3** in a cut view;
(6) FIG. **5** shows a cut view of a first primary stage piston and a second secondary stage piston attached together;
(7) FIG. **6** shows a more detailed cut view of FIG. **1** in the area of the first primary stage piston;
(8) FIG. **7** shows a detailed cut view of a drive arrangement;
(9) FIG. **8** shows a detail of a slot in the first sliding block guide;
(10) FIG. **9** shows a cut view of the first embodiment perpendicular to the cut view of FIG. **7**;
(11) FIG. **10** shows a detailed cut view of a drive arrangement;
(12) FIG. **11** shows a cut view of the second embodiment at a different angle than FIG. **10**;
(13) FIG. **12** shows a detailed cut view of a drive arrangement according to a third embodiment; and
(14) FIG. **13** shows a schematic view of a vehicle.
DETAILED DESCRIPTION
(15) The present disclosure provides a piston type pump which allows for reduced manufacturing cost, improved wear characteristics, compact design and/or low noise emissions.
(16) The present disclosure provides for a first sliding bush arranged between a first eccentric and a first sliding block guide, such that an outer surface of said first sliding bush corresponds to said inner surface of the first sliding block guide and that translational movement of said first sliding bush relative to said first sliding block guide is allowed along a main axis and restricted along a minor axis. The inner surface of said first sliding block guide forms a slot, wherein the first eccentric and the first sliding bush are located. The first sliding bush comprises an inner bore which is positioned on a corresponding outer surface of the first eccentric. Preferably, the first eccentric is rotatable in the first sliding bush about a rotational axis, which is parallel to a rotational axis of the drive shaft. A first direction of the first eccentric forms its central axis which is parallel to the rotational axis of the drive shaft und performs an orbital motion around said rotational axis when the drive shaft is driven. The first sliding bush is slidably seated in the first sliding block guide such that movement of the sliding bush relative to the first sliding block guide is allowed along the main

axis. The main axis of the sliding block guide is perpendicular to said rotational axis of the drive shaft. Preferably, the main axis is perpendicular to a cylinder axis of a first primary stage cylinder, which corresponds to the first primary stage piston. The minor axis of the first sliding block guide is perpendicular to the said rotational axis as well as the main axis. It is preferred that the minor axis is parallel or coaxial to the cylinder axis of the first primary stage cylinder. Movement of the first eccentric along the main axis results in a sliding motion of the eccentric and the first sliding bush along the inner surface of the sliding block guide. When the first eccentric and the first sliding bush move parallel to the minor axis a relative motion between the sliding guide block and the sliding bush is restricted and a motion of the first eccentric is transferred to the first sliding block guide via the first sliding bush. Thus, an orbital motion of the eccentric around the rotational axis of the drive shaft is converted to a rectilinear motion of the first sliding block guide parallel to the minor axis. The sliding bush allows for minimum friction between the first sliding block guide and the first sliding bush and/or between the first sliding bush and the first eccentric. A first material forming the first sliding bush is preferably one of carbon filled polymers, PTFE, ABS, PEEK, Nylon, PPS, Xylonite, fiber reinforced plastics, bronze, ceramics and/or white metal.

(17) Furthermore, the first material forming the first sliding bush can have a hardness that is lower than a hardness of a material forming the eccentric and/or the first sliding block guide. The first sliding bush can compensate for existing manufacturing deviations of the first sliding block guide and/or the first eccentric and thus allows for lower manufacturing costs of said components. Preferably the first sliding block guide and the first primary piston are integrally formed. For example, the first primary piston, the first sliding block guide, and a first piston rod connecting the first primary piston and the first sliding block guide can be casted in one piece. Casting production allows for low manufacturing costs at high quantities but in general reachable manufacturing tolerances are low. The rough tolerances can be compensated by the sliding bush and thus lower manufacturing costs can be achieved. It should be understood that the disclosure is not limited to casted first sliding block guides. Rough tolerances can also result from other manufacturing methods such as milling, turning, casting and/or additive manufacturing. In order to reduce manufacturing cost and time or to improve process stability rough tolerances are often accepted.

(18) Preferably, said inner surface of the first sliding block guide comprises a first inner wall parallel to said main axis and a second inner wall parallel to said main axis and spaced from said first inner wall by a slot width. A first dimension of the sliding bush perpendicular to the first direction, thus a dimension along the minor axis of the sliding block guide, is smaller or equal to said slot width such that the sliding bush can be placed in between the first and second inner wall. For example, the outer surface of the sliding bush can have a rectangular cross-section, wherein a first side of the sliding bush is parallel to said first inner wall of the sliding block guide and a second side of the sliding bush is parallel to said second inner wall of the sliding block guide. Preferably, the first inner wall and the second inner wall of said first sliding block guide are symmetrical to a plane perpendicular to the minor axis.

(19) In a preferred embodiment of the piston type pump said outer surface of the first sliding bush is tapered along a first direction parallel to a rotational axis of said drive shaft towards a first end of said drive shaft and said inner surface of the first sliding block is correspondingly tapered along the first direction towards said first end. Thus, the first sliding bush becomes gradually smaller along the first direction starting from an end of the drive shaft opposite to the first end towards said first end. Preferably a first dimension of the first sliding bush, which is measured parallel to the minor axis, gradually becomes smaller, while a second dimension of the first sliding bush, which is perpendicular to the first dimension and the first direction, may stay constant. Accordingly, a free inner surface or slot width of the first sliding block guide gradually becomes smaller starting form said opposite end of the drive shaft towards the first end of the drive shaft. Thus, a width of the slot, measured parallel to the minor axis gradually decreases along the first direction. A minimum width of the slot is larger than a diameter of the first eccentric. The inner surface of the first sliding

block guide and the outer surface of the first sliding bush are tapered correspondingly such that the inner surface of the first sliding block guide forms a negative of the corresponding outer surface of said first sliding bush. Preferably there is surface contact between said inner surface of the first sliding block guide and said outer surface of the first sliding bush and gaps are reduced or eliminated. Gaps often result in unwanted noise levels, increased wear and/or stability problems of the pump and therefore should be reduced or eliminated. The inner surface and/or outer surface can taper at a constant rate and/or can be variable. For example, the first sliding bush can be shaped as a truncated cone or bell wherein the inner surface of said first sliding block guide resembles a surface line of said cone or bell. In a particularly preferred embodiment the first sliding bush comprises four perpendicular sides forming the outer surface, wherein two opposing sides of said four sides are tapered and the two remaining sides are not tapered. Due to manufacturing tolerances an inner width of said inner surface of the first sliding block guide is variable and gaps between said first sliding bush and said first sliding block guide may exist. By tapering the inner and outer surface, areal contact between the first sliding bush and the first sliding block guide can be achieved by relative movement of the first sliding bush and the first sliding block guide parallel to the rotational axis of said drive shaft. Increased wear resulting from line contact as well as gaps can be avoided. A first axial length of said inner surface of the first sliding block guide, measured along the first direction, can be smaller than or equal to a corresponding second axial length of said outer surface of the first sliding bush. Preferably said first axial length is 50% to 100%, particularly preferred 70% to less than 100%, of said second axial length. Thus, the first sliding bush protrudes from the first inner surface at one or two sides.

(20) In another preferred embodiment the first sliding bush is biased towards said first end of said drive shaft by a first biasing member, such that said outer surface of the first sliding bush contacts said inner surface of the first sliding block guide. A first tapered end of the first sliding bush and a first tapered end of the first sliding block guide are oriented towards said first end of the drive shaft. Thus, the first sliding bush is continuously pushed into the inner surface or slot of the first sliding block guide and areal surface contact can be ensured. During operation of the pump the first sliding bush slides relative to the first sliding block guide and wear occurs on contacting surfaces of the first sliding bush and the first sliding block guide. The slot width of the first sliding block guide increases while at least a first dimension of the first sliding bush is decreased. Hence, gaps between the first sliding bush and the first sliding block guide are created. By biasing the tapered first sliding bush towards the tapered inner surface of said first sliding block guide, wear can be compensated and contact between the components can be ensured. The first sliding bush is pushed or pulled into the first sliding block guide along the first direction. Since wear is compensated, a movement of the first primary stage piston can be controlled effectively. The formation of gaps between the first sliding bush and the first sliding block be reduced or avoided. Such gaps generally lead to undesirable noise generation during pump operation. Furthermore, a travel of the piston during operation is constant, resulting in stabile working properties of the pump. Preferably the first tapered end of the first sliding bush is located adjacent to the first tapered end of the first sliding block guide. Preferably, the biasing member is spaced from said inner surface of the first sliding block guide in the first direction. As has been described above, a second axial length of the first sliding bush in said first direction can be larger than a first axial length of said first sliding block guide. Thus, the first sliding bush protrudes from the first inner surface on at least a first end. Preferably the biasing member is located on said first end.

(21) According to a further preferred embodiment said first biasing member applies a first biasing force in a range of larger 0 N to 40 N, preferably 5 N to 30 N, particularly preferred 12 N to 20 N, on said first sliding bush in said first direction.

(22) Moreover, it is preferred that a biasing force, applied by said first biasing member, is substantially symmetrical to said main axis. The amount of friction and wear between said outer surface of the first sliding bush and said inner surface of the first sliding block guide is at least

dependent on a contact area, material properties on the inner surface and the outer surface as well as the biasing force applied to the sliding bush. Due to the tapered shape, the biasing force leads to a reaction force perpendicular to the inner surface and/or outer surface. If the biasing force is chosen excessively wear on the first sliding bush and/or the first sliding block guide is increased. By choosing the biasing force in the preferred range, friction and/or wear between said inner surface and said outer surface can be minimized while allowing for constant wear compensation. By applying the biasing force symmetrically, symmetrical wear on the inner surface and/or outer surface can be ensured. Uneven wear might lead to local overloads and result in increased wear, functional loss and/or stability problems of the pump. Preferably, the biasing member pushes said first sliding bush into said first sliding block guide. However, it should be understood that the first biasing force can be a pulling force such that the first sliding bush is pulled into the first sliding block guide.
(23) In a further preferred development of the piston type pump said first biasing member is a spring clip attached to said first sliding block. Preferably, the spring clip comprises a first recess such that the first eccentric and/or the drive shaft can protrude through said recess. The spring clip comprises a biasing section contacting said first sliding bush, wherein the first sliding bush slides along the spring clip parallel to the main axis. It is preferred that the spring clip applies a uniform biasing force to the first sliding bush independently of a relative position of the first sliding bush in the first sliding block guide. Preferably the spring clip covers an opening of said first sliding block guide perpendicular to said inner surface. By attaching said spring clip to the first sliding block guide, a relative movement of the spring clip relative to the first slid block guide is inhibited. Since the spring clip is attached to the first sliding block guide it biases the first sliding bush towards said first end without applying torque on the first sliding block guide. Therefore, a rotation of the first primary stage piston in the first primary stage cylinder is avoided. It is preferred that the spring clip comprises a first hook section engaging a corresponding attachment section of said first sliding block guide. Preferably, the spring clip comprises a second hook section engaging a second corresponding attachment section of said first sliding guide block, wherein the first and second hook sections are formed on opposing ends of said spring clip such that the biasing section is located between the first and second hook section. The spring clip is fixed to the first sliding block guide via the first hook section and/or the second hook section. It is further preferred that the spring clip can be attached to the first sliding block guide in a one-hand operation and/or tool-less operation. By using a spring clip arrangement, a part count of the piston type pump can be reduced. It is also preferred that the spring clip prevents the first sliding bush from slipping out of a second end of said first sliding block guide along said first direction. It should be understood that other methods for fixing the spring clip to the first sliding block guide are also preferred. For example, the spring clip can be attached to the first sliding block guide by screwing, bolting, welding and/or bonding. In a particularly preferred embodiment the spring clip is integrally formed with the sliding block guide.
(24) According to a preferred embodiment said first biasing member is a wave spring, a flat coil, a Belleville washer and/or a wave washer connected to said first eccentric by a first retaining screw. The retaining screw is screwed into a corresponding internal thread of said first eccentric. Furthermore, one or multiple washers can be located between the wave spring, flat coil, Belleville washer and/or wave washer and the first sliding bush and/or between the wave spring, flat coil, Belleville washer and/or wave washer and a screw head of said retaining screw. It is further preferred that the wave spring, flat coil, Belleville washer and/or wave washer is connected to said first eccentric by a first threaded nut screwed to a corresponding external thread of the first eccentric. Again, one or more washers can be located between the biasing member and said threaded nut and/or between the biasing member and the first sliding bush. In this embodiment, the biasing member is not fixed to the first sliding block guide.
(25) Moreover, it is preferred that said first sliding bush comprises an end stop larger than a

maximum slot width of said inner surface. Said slot width is measured parallel to the minor axis of the first sliding block guide. Preferably, said slot width is constant in a contacting are of said inner surface, wherein the first sliding bush contacts the first sliding block guide in said contacting area. The end stop can be formed as a protrusion on the first sliding bush extending perpendicular to the first direction. The end stop prevents the first sliding bush from being pushed or pulled entirely through the first sliding block guide by the first biasing member. It is preferred that the first biasing member contacts said end stop. The end stop provides an increased surface area for contacting the biasing member. Surface pressure induced on the first sliding bush by the biasing force decreases and wear on the first sliding bush is reduced.

(26) In a further preferred embodiment said end stop is spaced from a corresponding stop face of said first sliding block guide. Preferably the corresponding stop face is perpendicular to said inner surface of the first sliding block guide. For example, the corresponding stop face and the end stop can be opposing flat surfaces. By spacing the corresponding stop face and the end stop, minimum wear during operation of the piston type pump can be ensured. When wear occurs on the inner surface of the first sliding block guide and the outer surface of the first sliding bush, the first sliding bush is pushed or pulled into the slot. A distance between the end stop and the corresponding stop face is reduced over time until the end stop and the corresponding stop face contact each other. Thus, emergency running characteristics can be ensured. It is preferred that the end stop is spaced from the corresponding stop face of said first sliding block guide by a range of larger 0% to 20% of a length of said inner surface, measured in the first direction. If the outer surface of the first sliding bush and/or the inner surface of the first sliding block guide is worn during operation of the pump, the first sliding bush is pushed or pulled into the first sliding block guide by the biasing member. The end stop is spaced apart from the corresponding stop face such that the first sliding bush can move relative to the first sliding block guide along the first direction. In a worn out state, the end stop contacts the corresponding stop face of said first sliding block guide and prevents the sliding bush from being pushed or pulled entirely through the first sliding block guide. A range of larger 0% to 20% of a length of said inner surface allows for wear compensation possibility and reduced friction while ensuring stable operation and compact design.

(27) Moreover, it is preferred that the drive shaft further comprises a second eccentric, the piston arrangement further comprises a second sliding block guide slidably seated on the second eccentric and wherein a second sliding bush is arranged between the second eccentric and the second sliding block guide. It should be understood that the second eccentric, the second sliding block guide and/or the second sliding bush can have similar properties as described above in respect to the first eccentric, the first sliding block guide and the first sliding block guide. Reference is made to above described features. Preferably the first eccentric and the second eccentric are phase-shifted by 180°.

(28) According to a particularly preferred embodiment, an outer surface of the second sliding bush facing the second sliding block guide is tapered along a second direction towards a second end of said drive shaft, an inner surface of the second sliding block guide corresponding to the outer surface of the second sliding bush is tapered along the second direction towards a second end of said drive shaft. The second end of said drive shaft and the first end of said drive shaft can be located on opposing sides or on the same side of said drive shaft. Preferably, the second sliding bush and the first sliding bush are identical or symmetrical and the first sliding block guide and the second sliding block guide are identical and or symmetrical. Thus, manufacturing cost can be reduced. Preferably, the second sliding bush is biased towards said second end of said drive shaft by a second biasing member, such that said outer surface of the second sliding bush contacts said inner surface of said second sliding block guide. The second biasing member preferably has features as described above for the first biasing member. However, the first biasing member can be formed as a spring clip, while the second biasing member is a wave spring, flat coil, Belleville washer and/or wave washer or vice versa. Forming them identical is particularly preferred to reduce costs.

(29) In a further preferred embodiment a tapered end of said first sliding bush faces a tapered end of said second sliding bush. A tapered end of a sliding bush is oriented in the first direction and reduced in thickness when compared to an opposite end. Preferably the tapered end of said first sliding bush and the tapered end of said second sliding bush are oriented towards a middle section of said drive shaft positioned between the first eccentric and the second eccentric. The first end of the drive shaft and the second end of the drive shaft are opposite ends of the drive shaft. If a first biasing member contacts the first sliding bush on a second end opposite to the tapered end, an assembly process can be improved, since the first biasing member can be assembled after the first sliding bush and the first sliding block guide. In an analogue manner if a second biasing member contacts the second sliding bush on a second end opposite to the tapered end, an assembly process can be improved, since the second biasing member can be assembled after the first sliding bush and the first sliding block guide. A first biasing force applied by a first biasing member and a second biasing force applied by a second biasing member are preferably oriented in opposite directions. Moreover, it is preferred that said first biasing force and said second biasing force are equal.
(30) Preferably the first sliding bush is rotationally fixed to the first sliding block guide. Furthermore, also a second sliding bush can be rotationally fixed to a second sliding block guide. The sliding motion between the first sliding block guide and the first sliding bush and/or between the second sliding block guide and the second sliding bush can be limited to a linear sliding motion. For example, an outer cross-section of the first sliding bush can be substantially rectangular, such that a longer side of the outer cross-section of the first sliding bush is larger than a smaller side of the inner cross-section of the first sliding block guide and/or wherein an inner cross section of the second sliding block guide can be substantially rectangular and a corresponding outer cross-section of the second sliding bush is substantially rectangular such that a longer side of the outer cross-section of the second sliding bush is larger than a smaller side of the inner cross-section of the second sliding block guide.
(31) According to further preferred embodiment the piston arrangement further comprises a first primary stage cylinder formed in the pump housing, in which said first primary stage piston is slidably seated, and a first secondary stage piston being slidably seated in a first secondary stage cylinder formed in the first primary stage piston. Preferably, the first primary stage piston and the first secondary stage cylinder are integrally formed. For example, the first secondary stage cylinder is machined in the first primary stage piston. They are, thus, preferably formed in a one-piece construction. Due to this arrangement, the overall size of the pump can be reduced. The second stage is formed inside the first stage and not adjacent to it or at any other position. While the first primary stage piston moves relative to the pump housing inside a first primary stage cylinder formed inside the piston housing, the first secondary stage piston moves inside the first primary stage piston.
(32) According to a further preferred embodiment the piston type pump is formed as a so-called twin piston type pump and therefore comprises a second primary stage piston and a second secondary stage piston, wherein the second primary stage piston is slidably seated in a second primary stage cylinder formed in the pump housing, and the second secondary stage piston is slidably seated in a second secondary stage cylinder formed in the second primary stage piston. Depending on how the different cylinders communicate with each other the second primary stage piston may also form a first tertiary stage piston and the second secondary stage piston may form a first quaternary stage piston. In this manner, the piston type pump, which, according to this embodiment, in total includes four pistons, can form a four-stage piston pump. However, particularly preferred is a two stage twin pump which includes two stages, with four pistons and thus a first and second first stage and a first and second secondary stage. As it has been described with respect to the first primary stage piston and the first secondary stage cylinder, also the second primary stage piston and the second secondary stage cylinder preferably are integrally formed, in particular preferred as a one-piece. It is further preferred that the minor axis of the first sliding

block guide and the minor axis of the second sliding block guide are parallel to each other. Moreover, it is preferred that a main axis of the first primary stage cylinder and the second primary stage cylinder are coaxial such that the piston type pump is a boxer type pump. Preferably the first secondary stage piston is attached to the second primary stage piston and the second secondary stage piston is attached to the first primary stage piston. Further it is particularly preferred that the first secondary stage piston and the second primary stage piston are integrally formed and the second secondary stage piston and the first primary stage piston are integrally formed.
(33) According to a second aspect, a piston type pump is provided, the piston type pump comprising a pump housing having at least one pump inlet and one pump outlet and a piston arrangement connected to a drive shaft which, when driven, sets the piston arrangement into movement, wherein the drive shaft comprises a first eccentric, and the piston arrangement comprises a first primary stage piston connected to a first sliding block guide slidably seated on the first eccentric, said first sliding block guide having a main axis and a minor axis characterized in that a first rolling bearing is being arranged between the first eccentric and the first sliding block guide, such that translational movement of said first rolling bearing relative to said first sliding block guide is allowed along the main axis and restricted along the minor axis, wherein the piston arrangement further comprises a first primary stage cylinder formed in the pump housing, in which said first primary stage piston is slidably seated, and a first secondary stage piston being slidably seated in a first secondary stage cylinder formed in the first primary stage piston.
(34) According to a third aspect, a vehicle is provided, in particular a passenger car, comprising a piston type pump according to any of the aforementioned preferred embodiments of a piston type pump according to the first aspect or according to the second aspect.
(35) It shall be understood that the pump may also be used in applications other than vehicles, and in particular other than braking systems. Other uses of pumps for generating a vacuum on a vehicle can include engine mounts, compressor waste-gate and bypass valves actuation. This type of pump could also feasibly be used to evacuate a housing for a KERS (Kinetic Energy Recovery System) for example. Furthermore, the pump can be used as a diagnostic pump for an automotive Evaporative Emissions Circuit (EVAP).
(36) For a more complete understanding of the present disclosure, the embodiments of the present disclosure will now be described in detail with reference to the accompanying drawings. The detailed description will illustrate and describe what is considered as a preferred embodiment. It should of course be understood that various modifications and changes in form or detail could readily be made without departing from the spirit of the present disclosure. It is therefore intended that the present disclosure may not be limited to the exact form and detail shown and described herein. The wording "comprising" does not exclude other elements or steps. The word "a" or "an" does not exclude the plurality. The wording "a number of" items comprising also the number 1, i.e. a single item, and further numbers like 2, 3, 4 and so forth. In the accompanying drawings:
(37) A piston type pump **1** according to the present disclosure is suitable to be mounted within a vehicle **100** (see FIG. **12**) and used as a vacuum pump to provide vacuum for a braking system or any other consumer in this vehicle. The piston type pump **1** in particular is suitable to be driven by an electric motor which for simplicity is not shown in the drawings.
(38) FIG. **1** shows the piston type pump **1** prepared to be used as a vacuum pump and to induce a vacuum at a pump inlet **4**. However, the same construction may also be used as a compressor. The embodiment shown in FIG. **1** to **6** is used to describe possible features of a piston type pump. Possible drive arrangements for a piston type pump are described with reference to FIG. **7** to **11**. It should be understood that all combinations of the drive arrangement according to the embodiments shown in FIG. **7** to FIG. **11** with the features of a piston type pump presented with reference to FIG. **1** to **6** are preferred. Furthermore, a piston type pump having different drive arrangements as described according to the embodiments shown in **7** to FIG. **11** for different pistons is preferred.
(39) The piston type pump **1** comprises a pump housing **2** which in the embodiment shown in FIG.

1 substantially is cylindrical. The pump housing **2** has a pump inlet **4** which can be connected to a consumer. Moreover, the pump housing comprises a pump outlet **6** which opens into the environment. The pump outlet **6** is formed as a simple opening in the pump housing **2**. Fluid, in particular air, which is drawn away from the pump inlet **4**, is not used and only discharged to the environment instead of being provided to any consumer, when the piston type pump **1** is used as a vacuum pump. Within the pump housing **2** a piston arrangement **8** is provided, which will be described in more detail below. The piston arrangement **8** is connected to a drive shaft **10** which, when driven, sets the piston arrangement **8** into movement for inducing a vacuum at the pump inlet in this embodiment. The drive shaft **10** is rotatable about a rotational axis A and may be connected to an electric motor.

(40) The piston arrangement **8** according to the embodiment shown in FIG. **1** comprises a first primary stage piston **12** which is slidably seated in a first primary stage cylinder **14** formed in the pump housing **2**. The first primary stage piston **12** in FIG. **1** is shown in its first end position, which is the position furthest away from the rotational axis A, however might travel within the first primary stage cylinder **14** to the left-hand side direction with respect to FIG. **1**, thus closer to the rotational axis A.

(41) The piston type pump **1** according to the shown embodiment is formed as a twin type two-stage piston pump and therefore also comprises a first secondary stage piston **16**, which is provided within a first secondary stage cylinder **18**, which is formed within the first primary stage piston **12**. The first primary stage piston **12**, therefore, is formed in a hollow manner, to form the first secondary stage cylinder **18**. The first primary stage piston **12** comprises a first primary stage piston wall **13** which defines the first secondary stage cylinder **18**. The first secondary stage cylinder **18** in particular is formed by an inner circumferential surface of the first primary stage piston wall **13** within the first primary stage piston **12**.

(42) In a similar fashion the piston arrangement **8** according to this embodiment also comprises a second primary stage piston **40**, which is slidably seated in a second primary stage cylinder **42**, which again is formed inside the pump housing **2**. The complete interior **3** of the pump housing **2** can be formed as a cylindrical hollow portion to form both, the first primary stage cylinder **14** and the second primary stage cylinder **42**.

(43) Also a second secondary stage cylinder **44** is provided which is slidably seated in a second secondary stage cylinder **46** formed within the second primary stage piston **40**. Again the second primary stage piston **40** comprises a second primary stage piston wall **41** which defines the second secondary stage cylinder **46** by its inner circumferential surface restricting a second hollow space **47**.

(44) Moreover, in FIG. **1** it can be seen that the first primary stage cylinder **14** comprises a first central axis B**1** and the second primary stage cylinder **42** comprises a second central axis B**2**, which are coaxial. Thus, the first and the second central axes B**1**, B**2** form a single axis on which the first primary stage piston **12** and the second primary stage piston **40** move. When the first secondary stage cylinder **18** and the second secondary stage cylinder **46** are formed concentrically within the respective first primary stage piston **12** and the second primary stage piston **40**, also the first secondary stage piston **16** and the second secondary stage piston **44** move coaxially with the first and second central axes B**1**, B**2**. Thus, the overall design of the piston type pump **1** is a boxer type piston type pump in which the single pistons move in opposing directions. This may lead to a well-balanced design.

(45) The first secondary stage piston **16** comprises a first piston rod **54** which extends through a first assembly opening **60** in the first primary stage piston wall **13**. The portion of the hollow space **19** which is on the opposite side of the piston rod **54** with respect to the first secondary piston face **30** can be named the first secondary stage working chamber. In the same manner the second secondary stage piston **44** comprises a second piston rod **56** which extends through a second assembly opening **58** formed in the second primary stage piston wall **41** of the second primary

stage piston **40**.

(46) Now beginning with FIG. **6**, the flow of fluid will be described in more detail.

(47) The pump inlet **4** (see FIG. **6**) is here only shown as a single opening which is in fluid connection with a first conduit **74** formed in the pump housing **2**. The conduit **74** is surrounded by a protuberance **75** of the pump housing **2**, which can be seen in FIG. **1** also. This first conduit **74** substantially extends in a parallel way to the first and second central axes B**1**, B**2**. The first conduit **74** on the one hand leads to a second conduit **76** formed in a first housing lid **78** which closes the pump housing **2** and also closes the first primary stage cylinder **14**. This second conduit **76** terminates in a first inlet chamber **80** which is closed to the environment by means of a first chamber lid **82**. The first inlet chamber **80** comprises a first inlet check valve **84** which allows fluid to flow through the first conduit **74**, the second conduit **76**, the inlet chamber **80** into the first primary stage cylinder **14**, but not vice versa. This is indicated by the arrows in FIG. **6**. The first inlet check valve **84** can be formed as a leaf valve and comprises a leaf **86** which is flexible and might be formed out of any flexible material, such as thin metal, elastomer or the like.

(48) A similar arrangement is provided on the other end of the pump housing **2** (see FIG. **1**). Even though FIG. **1** is not as detailed as FIG. **6**, it shall be understood that the same arrangement is provided. In particular, the pump housing **2** comprises a second housing lid **88** comprising a second inlet chamber **90** with a second inlet check valve **92** and a respective second leaf of the second inlet check valve **92**. A third conduit **96** is provided in the second housing lid **88**, however, not shown in cut view in FIG. **1**, but connected to the first conduit **74** in a similar manner as it has been described with respect to the second conduit **76**. Again the second inlet check valve **52** allows fluid to enter the second stage cylinder **46** through the first conduit **74**, the third conduit **96**, the second inlet chamber **90** and the second inlet check valve **92**. The first housing lid **78** and the second housing lid **88** may be formed identical to each other or in a mirrored fashion. In any case manufacturing of the piston type pump **1** is simplified.

(49) When now the drive shaft **10** begins to rotate due to operation of an electric motor attached to the drive shaft **10**, the first and second primary stage pistons **12**, **40** (see FIG. **1**) will move along the respective first and second central axes B**1**, B**2** toward the rotational axis A. Thus, the working chamber, which is formed between the piston housing **2**, the respective housing lids **78**, **88** and the respective first and second primary stage pistons **12**, **40** will be enlarged and therefore fluid will be drawn through the pump inlet **4**, the first conduit **74**, the second and third conduits **76**, **96**, the first and the second inlet chambers **80**, **90** and the first and second inlet check valves **84**, **92**. Due to the movement of the first and second primary stage pistons **12**, **40** the primary stage vacuum is induced at the pump inlet **4**.

(50) When now the drive shaft **10** continues to rotate, the first and second primary stage pistons **12**, **40** will again be pushed outwardly, i.e. away from the rotational axis A. The respective first and second working chambers will become smaller and residual fluid, which is in these working chambers, will be compressed. The first and second inlet check valves **84**, **92** prevent this fluid from flowing toward the pump inlet **4** again. However, this fluid needs to exit the piston type pump **1**. To achieve this, the first primary piston face **24** is provided with a first primary outlet **26** which in turn is provided with a first primary check valve **28**. Thus, the fluid contained in the first working chamber can flow through the first primary outlet **26** and the first primary check valve **28** into the first secondary stage cylinder **18**.

(51) In the same manner also a second primary piston face **48** of the second primary stage piston **40** is provided with a second primary outlet **50** which in turn is provided with a second primary check valve **52**. Thus, fluid contained in the second working chamber may flow through the second primary outlet **50** and the second primary check valve **52** into the second secondary stage cylinder **46** upon movement of the second primary piston **40** away from the rotational axis A.

(52) Both, the first and second primary check valves **28**, **52** again might be formed as leaf valves and comprise respective first and second primary check valve leaves **96**, **98** which can be identical

to leaves **86**, **94**.
(53) For a more easy manufacturing and assembly, the first primary stage piston face **24** is defined by a first primary stage piston lid **70** attached to the first primary piston wall **13**. This first primary stage piston lid **70** carries the first primary check valve **28**. Also the second primary stage piston face **48** is defined by a second primary stage piston lid **72** attached to the second primary piston wall **41**. This second primary stage piston lid **72** carries the second primary check valve **52**.
(54) When the first and second primary stage pistons **12**, **40** are in the central position, thus proximal to the rotational axis A, the first and second secondary stage pistons **16**, **44** are at the outermost position, thus most distal to the rotational axis A, due to their connection to the first and second eccentrics **20**, **21**. In this position the first and second secondary stage pistons **16**, **44** are proximal to the first and second primary check valves **28**, **52** and the respective working chamber is small. Upon rotation of the central drive shaft **10** and movement of the first and second primary stage pistons **12**, **40** outwardly, the first and second secondary stage pistons **16**, **44** are drawn inwardly toward the rotational axis A, therefore enlarging the respective first and second secondary stage working chambers. A vacuum is induced and additional fluid may flow from the pump inlet **4** through the first and second inlet check valves **84**, **92**, the first and second primary check valves **28**, **52** into the first and second secondary stage working chambers.
(55) On the other hand, when the drive shaft **10** rotates further, the first and second secondary stage pistons **16**, **44** are pushed outwardly again, thus decreasing the respective first and second secondary stage working chambers. The fluid, contained in these first and second secondary stage working chambers needs to exit the piston type pump **1**.
(56) To achieve this, the first secondary piston face **30** is provided with a first secondary outlet **32**, which in turn is provided with a first secondary check valve **34** (see FIGS. **3**, **4** and **6**). As shown in FIG. **6**, fluid can pass through this first secondary check valve **34** and out of the pump outlet **6**.
(57) In the same manner, also the second secondary stage cylinder **46** is provided with a second secondary outlet **49** in a second secondary piston face **45** and a second secondary check valve **51**. Again, fluid may pass through this second secondary check valve **51** and out of the pump outlet **6**.
(58) Afterwards, the drive shaft **10** rotates further and again moves the first and second secondary stage pistons **16**, **44** toward the rotational axis A. It shall be understood that dependent on how the first, second and third conduits **74**, **76**, **96** are arranged, also the, for example, first secondary outlet **32** may be guided into the second primary stage working chamber, thus into the second primary stage cylinder **42** and the vacuum may be further decreased. In such an arrangement the piston type pump **1** would be a four stage vacuum pump instead of a two stage twin type vacuum pump as shown in the embodiments in the attached figures.
(59) Beginning with FIG. **7** the drive assembly according to a first embodiment is explained in more detail. The first primary stage piston **12** is connected to a first sliding block guide **62** seated on a first eccentric **20** of the drive shaft **10**. The first eccentric **20** is integrally formed with the drive shaft **10** comprising a first eccentricity e**1**, measured with respect to the rotational axis A of the drive shaft **10**. In this embodiment the first eccentric **20** is formed as a crank pin **112** having a circular cross section. An inner surface **114** of the first sliding block guide **62** forms a slot **116** in the sliding block guide. A first sliding bush **118** is arranged between the first eccentric **20** and the first sliding block guide **62**. An outer surface **120** of the first sliding bush **118** is directed towards said inner surface **114** of the first sliding block guide **62**. In an analogous manner an outer surface **121** of the second sliding bush **119** is directed towards an inner surface **115** of the second sliding block guide **64**. Furthermore, the first sliding bush **118** comprises an inner bore **122** for receiving the first eccentric **20**.
(60) A minor axis C**2** of the first sliding block guide **62** is perpendicular to a main axis C**1** (FIG. **8**) and to a first direction D**1** (which is perpendicular to the plane of the drawing in FIG. **8**). The inner surface **114** comprises a first inner wall **124** parallel to the main axis C**1** and a second inner wall **126** parallel to said first inner wall **124**. Preferably, the first inner wall **124** and the second inner

wall **126** are symmetrical to a first plane S**1** which is defined by the main axis C**2** and the first direction D**1**. The first inner wall **124** and the second inner wall **126** are spaced from each other at a slot width SW measured parallel to the minor axis C**2**. A slot height SH of the slot **116** measured parallel to the main axis C**1** is preferably chosen such that the first sliding bush **118** does not contact a third inner wall **128** and a fourth inner wall perpendicular to the first inner wall **124** and the second inner wall **126**. Corners **132** of the slot **116** can be rounded or angular. Also, the third and fourth walls **128**, **130** may be curved.

(61) According to this embodiment, the inner surface **114** of the first sliding block guide **62** and the outer surface **120** of the first sliding bush **118** are tapered along the first direction D**1** towards a first end **134** of the drive shaft **10** (FIG. **7**). Here the second sliding bush **119** is tapered towards a second end **135** of the drive shaft **10** in an analogous manner. The inner surface **114** substantially forms a negative of the outer surface **120**, such that the first sliding bush **118** contacts the inner surface **114** along the first direction D**1**. A tapered end **136** of the first sliding bush **118** is located adjacent to a tapered end **138** of the slot **116**. An opposite end **140** of the first sliding bush **118**, opposite to the tapered end **136**, protrudes from the inner surface **114**. The outer surface **120** of the first sliding bush **118** is substantially bell shaped in a first cross-section perpendicular to the main axis C**1** such that said outer surface **120** is variably tapered in the first direction D**1**. Preferably, an outer width OW of the first sliding bush **118** is essentially equal to the corresponding slot width SW of the slot **116** along the first direction D**1**. In a second cross-section perpendicular to the first cross-section the first sliding bush **118** is preferably rectangular such that surfaces of the first sliding bush which are perpendicular to the main axis C**1** are parallel to the first direction D**1**.

(62) A first biasing member **142** applies a biasing force F**1** to the first sliding bush **118**. Here the first biasing member **142** is formed as a spring clip **144**. The spring clip **144** applies the biasing force F**1** to an end stop **146** of the first sliding bush **118**, which is positioned at the opposite end **140** of the first sliding bush **118**. Preferably, the biasing force F**1** is parallel to the first direction D**1** such that the first sliding bush **118** is pushed into the slot **116**. It is further preferred that the biasing force F**1** is applied equally to the first sliding bush **118** such that a first reaction force F**2** on the first inner wall **124** and a second reaction force F**3** on the second inner wall **126** are equal. The end stop **146** of the first sliding bush **118** is spaced from a stop face **148** of the first sliding block guide **62** in the first direction D**1**. The end stop **146** is formed as a circumferential protrusion **150** or bead of the first sliding bush **118** which has an end stop width ESW that is larger than a maximum slot width MSW of the slot **116** (FIG. **7**). Thus, the first sliding bush **118** is prevented from being entirely pushed through the slot **116** when the outer surface **120** and/or the inner surface **114** are worn.

(63) The spring clip **144** comprises a first hook section **152** engaging a first attachment section **154** of the first sliding block guide **62** and a second hook section **156** engaging a second attachment section **158** of the first sliding block guide **62**. The first hook section **152** and the second hook section **156** are arranged such that even when the biasing force F**1** is applied by the spring clip **144**, the spring clip **144** is fixed to the first sliding block guide **62**. In this embodiment the first attachment section **154** and the second attachment section **158** are formed as planes perpendicular to the first direction D**1**. Preferably, the first hook section **152** and the second hook section **156** are formed on opposing ends of the spring clip **144** while a biasing section **160** is positioned in between the hook sections **152**, **156**. The biasing section **160** contacts the end stop **146** of the first sliding bush **118**. Furthermore, the biasing section **160** covers the slot **116** such that the first sliding bush **118** is prevented from slipping out of the slot **116**.

(64) When the drive shaft **10** is rotated about the rotational axis A, the first eccentric **20** performs an orbital movement around said rotational axis A. Since the first sliding bush **118** is located on the first eccentric **20** said orbital movement is transferred to said first sliding bush **118**. The first sliding bush **118** is rotationally fixed within the first sliding block guide **62** such that the first eccentric **20** rotates relative to first sliding bush **118** in the inner bore **122** and only translational movement is transferred to the first sliding block guide **62**. The orbital movement comprises a first component

parallel to the main axis C**1** and a second component parallel to the minor axis C**2**. It should be understood that since the first eccentric **20** performs an orbital movement, movements parallel to the main axis C**1** and parallel to the minor axis C**2** occur simultaneously. If a component of the orbital movement along the main axis C**1** is transferred to the first sliding bush **118**, said first sliding bush **118** slides relative to the first sliding block guide **62** along the main axis C**1** and no movement is transferred to the first sliding block guide **62**. Since the outer surface **120** of the first sliding bush **118** contacts the first inner wall **124** and the second inner wall **126** of the first sliding block guide **62**, translational movement of said first sliding bush **118** relative to said first sliding block guide **62** is not possible along the minor axis C**2**. Thus, the component of said orbital movement of the first eccentric **20** parallel to the minor axis C**2** is transferred to the first sliding block guide **62** via the first sliding bush **118**. The orbital movement of the first eccentric **20** is transformed into a rectilinear movement of the first sliding block guide **62** along its minor axis C**2**.
(65) In a second embodiment (see FIG. **10**) the first sliding bush **118** is biased towards the first sliding block guide **62** by a wave spring **162**. Again, the outer surface **120** (not shown in FIG. **10**) of the first sliding bush **118** and the inner surface **114** of the first sliding block guide **62** are tapered in the first direction D**1**. An upper surface **164** and a lower surface **166** of the outer surface **120** of said sliding bush **118** are parallel to each other and to the first direction D**1**. The wave spring **162** is positioned at the end stop **146** and evenly applies the biasing force F**1**. First eccentric **20** comprises an internal thread **168** which corresponds to a retaining screw **170**. Said retaining screw **170** is screwed into the internal thread **168** and fixes the wave spring **162** to the first eccentric **20**. A thrust washer **172** and a slip washer **174** are arranged between a screw head **176** of said retaining screw **170** and the wave spring **162**. The thrust washer **172** is positioned adjacent to the screw head **176** and has a larger outer diameter than said screw head **176**. Using a thrust washer **172** it is possible to use a standardized screw as retaining screw **170** while allowing for adequate material thickness of the first eccentric **20**. The wave spring **162** acts against the thrust washer **172** and imparts a torque T (indicated by arrows in FIG. **11**) to the first sliding block guide **62** via the first sliding bush **118**. Such a torque is absent in the arrangement according to the first embodiment as presented in FIG. **7** to FIG. **9**.
(66) During operation the first eccentric **20** rotates inside the inner bore **122** of the first sliding bush **118** while the retaining screw **170** is fixed to the first eccentric **20**. Relative motion between a first opposing end **178** and a second opposing end **180** of the wave spring **162** might cause damage to said wave spring **162** and/or change the biasing force F**1**. The slip washer **174** allows for relative rotational movement between said slip washer **174** and the thrust washer **172** about and parallel to the first direction D**1**. During operation the screw head **176** and the thrust washer **172** rotate in the first sliding bush while the slip washer **174** slips relative to said thrust washer **172**. Thus, torsional forces on the wave spring **162** are avoided.
(67) It should be understood that the second eccentric **22**, second sliding block guide **64** and a second sliding bush **119** can have similar features as described regarding the first eccentric **20**, the first sliding block guide **62** and the first sliding bush **118**.
(68) With reference to FIG. **12** a third embodiment of a piston type pump **1** is described. Elements that are similar or identical to other embodiments have the same reference numerals used in the previous figures. Again, all features described with reference to FIG. **1** to FIG. **6** can be present in this embodiment. The drive shaft **10** comprises a first eccentric **20** which is located in a slot **116** of a first sliding block guide **62**. In this embodiment a rolling bearing **182** is located between the first eccentric **20** and the first sliding block **62**. It shall be understood that the rolling bearing **182** can be one of: roller bearing, needle roller bearing, barrel roller bearing and/or any other type of rolling bearing. An inner ring **184** of the rolling bearing **182** is fixed to the first eccentric **20** via a fixing member **186**. Preferably, the fixing member **186** is screwed to the first eccentric **20** via a fixing screw (not shown in FIG. **12**). For fixing the inner ring **184** of the rolling bearing **182** the first eccentric **20** comprises a first bearing stop face **188** and the fixing member **186** comprises a second

bearing stop face **190**. The first bearing stop face **188** protrudes perpendicular to the first direction D**1**. In an analogous manner the second bearing stop face **188** protrudes perpendicular to the first direction D**1**. The inner ring **184** of the rolling bearing **182** is fixed between the first bearing stop face **188** and the second bearing stop face **190**. An outer ring **192** of the rolling bearing **182** is positioned adjacent to the inner surface **114** of the first sliding block guide **62** and slidable to said inner surface **114**. During operation drive shaft **10** is driven and eccentric **20** as well as the inner ring **184** of said rolling bearing **182** perform an orbital movement around the rotational axis A. The outer ring **192** is rotatable supported relative to the inner ring **184** around the first direction D**1** by multiple rollers **194** (shown schematically in FIG. **12**). Thus, wear on the first eccentric **20** can be reduced. With respect to FIG. **12** the outer ring **192** of the rolling bearing **182** slides up and down in said slot **116** of the first sliding block guide **62**. A motion of the first eccentric **20** along the minor axis C**2** is transferred to the first sliding block guide by the rolling bearing **182** such that a pendulum motion of the first primary stage piston **12** is eliminated. Preferably, the outer ring **192** of said rolling bearing **182** and/or the inner surface **114** of said sliding block guide **62** are made of a highly wear resistant material. Preferably, the inner surface **114** and/or the outer ring **192** are hardened.

(69) FIG. **13** now depicts a schematic drawing of a vehicle **100**. Vehicle **100** preferably is formed as a passenger car, or a light truck and comprises a pneumatic braking system **102**. The braking system **102** is shown by lines **104** leading to wheels **106***a*, **106***b*, **106***c*, **106***d* for providing the wheels **106***a*, **106***b*, **106***c*, **106***d* with the respective braking pressure. Lines **104** are connected to a central module **108**. The vehicle **100** moreover comprises an engine **110** and a piston type pump **1**, which is herein used as a vacuum pump **1**. piston type pump **1** provides the braking system **102** with vacuum, which e.g. could be used by a brake booster of the braking system **102**, which could be implemented in the central module **108**.

(70) While subject matter of the present disclosure has been illustrated and described in detail in the drawings and foregoing description, such illustration and description are to be considered illustrative or exemplary and not restrictive. Any statement made herein characterizing the invention is also to be considered illustrative or exemplary and not restrictive as the invention is defined by the claims. It will be understood that changes and modifications may be made, by those of ordinary skill in the art, within the scope of the following claims, which may include any combination of features from different embodiments described above.

(71) The terms used in the claims should be construed to have the broadest reasonable interpretation consistent with the foregoing description. For example, the use of the article “a” or “the” in introducing an element should not be interpreted as being exclusive of a plurality of elements. Likewise, the recitation of “or” should be interpreted as being inclusive, such that the recitation of “A or B” is not exclusive of “A and B,” unless it is clear from the context or the foregoing description that only one of A and B is intended. Further, the recitation of “at least one of A, B and C” should be interpreted as one or more of a group of elements consisting of A, B and C, and should not be interpreted as requiring at least one of each of the listed elements A, B and C, regardless of whether A, B and C are related as categories or otherwise. Moreover, the recitation of “A, B and/or C” or “at least one of A, B or C” should be interpreted as including any singular entity from the listed elements, e.g., A, any subset from the listed elements, e.g., A and B, or the entire list of elements A, B and C.

LIST OF REFERENCE CHARACTERS

(72) **1** piston type pump **2** pump housing **3** interior **4** pump inlet **6** pump outlet **8** piston arrangement **10** drive shaft **12** first primary stage piston **13** first primary stage piston wall **14** first primary stage cylinder **16** first secondary stage piston **18** first secondary stage cylinder **19** hollow space **20** first eccentric **22** second eccentric **24** first primary piston face **26** first primary outlet **28** first primary check valve **30** first secondary piston face **32** first secondary stage piston outlet **34** first secondary check valve **40** second primary stage piston **41** second primary stage piston wall **42**

second primary stage cylinder **44** second secondary stage piston **45** second secondary piston face **46** second secondary stage cylinder **47** second hollow space **48** second primary piston face **49** second secondary outlet **50** second primary outlet **51** second secondary check valve **52** second primary check valve **54** first piston rod **56** second piston rod **58** second assembly opening **60** first assembly opening **62** first sliding block guide **64** second sliding block guide **70** first piston lid **72** second primary stage piston lid **74** first conduit **75** protuberance **76** second conduit **78** first housing lid **80** first inlet chamber **82** first chamber lid **84** first inlet check valve **86**, **94** leaf **88** second housing lid **90** second inlet chamber **92** second inlet check valve **96** third conduit **100** vehicle **102** pneumatic braking system **104** lines **106***a*, **106***b*, **106***c*, wheels **108** central module **110** engine **112** crank pin **114** inner surface **115** inner surface (second sliding block **116** slot **118** first sliding bush **119** second sliding bush **120** outer surface **121** outer surface (second sliding bush) **122** inner bore **124** first inner wall **126** second inner wall **128** third inner wall **130** fourth inner wall **132** corner **134** first end (drive shaft) **135** second end (drive shaft) **136** tapered end (first sliding bush) **138** tapered end (slot) **140** opposite end **142** first biasing member **144** spring clip **146** end stop **148** stop face **150** protrusion **152** first hook section **154** first attachment section **156** second hook section **158** second attachment section **160** biasing section **162** wave spring **164** upper surface **166** lower surface **168** internal thread **170** retaining screw **172** thrust washer **174** slip washer **176** screw head **178** first opposing end **180** second opposing end **182** rolling bearing **184** inner ring **186** fixing member **188** first bearing stop face **190** second bearing stop face **192** outer ring **194** roller e**1** first eccentricity e**2** second eccentricity A rotational axis B**1** first central axis B**2** second central axis C**1** main axis C**2** minor axis D**1** first direction ESW end stop width F**1** biasing force F**2** first reaction force F**3** second reaction force MSW maximum slot width OW outer width S**1** first plane SW slot width SH slot height T torque

Claims

1. A piston type pump, comprising: a pump housing having a pump inlet, a pump outlet, and a piston arrangement connected to a drive shaft, the drive shaft being configured to drive the piston arrangement, the drive shaft comprising a first eccentric, the piston arrangement having a first primary stage piston connected to a first sliding block guide slidably seated on the first eccentric, the first sliding block guide having a main axis, a minor axis, and an inner surface, wherein a first sliding bush is arranged between the first eccentric and the first sliding block guide, wherein an outer surface of the first sliding bush corresponds to the inner surface of the first sliding block guide and a translational movement of the first sliding bush relative to the first sliding block guide is allowed along the main axis and restricted along the minor axis, wherein the outer surface of the first sliding bush is tapered along a first direction that is parallel to a rotational axis of the drive shaft towards a first end of the drive shaft, and wherein the inner surface of the first sliding block guide is correspondingly tapered along the first direction towards the first end of the drive shaft.
2. The piston type pump according to claim 1, wherein the first sliding bush is biased towards the first end of the drive shaft by a first biasing member such that the outer surface of the first sliding bush contacts the inner surface of the first sliding block guide.
3. The piston type pump according to claim 2, wherein the first biasing member applies a first biasing force in a range of larger 0 N to 40 N on the first sliding bush in the first direction that is parallel to the rotational axis of the drive shaft.
4. The piston type pump according to claim 2, wherein the first biasing member is a spring clip attached to the first sliding block guide.
5. The piston type pump according to claim 2, wherein the first biasing member is one of a wave spring, a flat coil, a Belleville washer and/or a wave washer that is connected to the first eccentric by a first retaining screw.
6. The piston type pump according to claim 2, wherein the first sliding bush comprises an end stop

that is larger than a maximum slot width of the inner surface of the first sliding block guide.
7. The piston type pump according to claim 6, wherein the end stop is spaced from a corresponding stop face of the first sliding block guide.
8. The piston type pump according to claim 1, wherein the drive shaft further comprises a second eccentric, wherein the piston arrangement further comprises a second sliding block guide slidably seated on the second eccentric, and wherein a second sliding bush is arranged between the second eccentric and the second sliding block guide.
9. The piston type pump according to claim 8, wherein an outer surface of the second sliding bush facing the second sliding block guide is tapered along a second direction towards a second end of the drive shaft, and wherein an inner surface of the second sliding block guide corresponding to the outer surface of the second sliding bush is tapered along the second direction towards a second end of the drive shaft.
10. The piston type pump according to claim 8, wherein a tapered end of the first sliding bush faces a tapered end of the second sliding bush.
11. The piston type pump according to claim 1, wherein the first sliding bush is rotationally fixed to the first sliding block guide.
12. The piston type pump according to claim 1, wherein the piston arrangement further comprises a first primary stage cylinder formed in the pump housing, wherein the first primary stage piston is slidably seated in the first primary stage cylinder, and a first secondary stage piston being slidably seated in a first secondary stage cylinder formed in the first primary stage piston.
13. A vehicle comprising the piston type pump according to claim 1.